

ECEN-629: Homework 2

Due on September 30, 2025 at 11:59 pm

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Problem 1

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is concave, and $a, b \in \text{dom } f$ with $a < b$.

(a) Prove the following inequality:

$$\frac{f(x) - f(a)}{x - a} \geq \frac{f(b) - f(a)}{b - a} \geq \frac{f(b) - f(x)}{b - x}.$$

for any $x \in (a, b)$.

(b) For $f = -x^2 + 2x + 5$, $a = 0$, and $b = 3$, draw a sketch to illustrate the inequalities above.

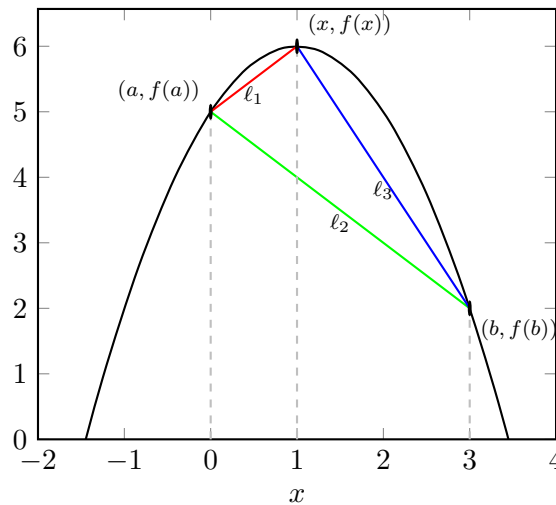
Solution

With $x \in (a, b)$, we can write $x = \theta a + (1 - \theta)b$ which gives $\theta = (x - b)/(a - b)$ for $a \neq b$.

$$\begin{aligned} f(\theta a + (1 - \theta)b) &\geq \theta f(a) + (1 - \theta)f(b) \\ f(x) &\geq \frac{x - b}{a - b}f(a) + \frac{a - x}{a - b}f(b) \\ f(x) - f(a) &\geq \left(\frac{b - x}{b - a} - 1\right)f(a) + \frac{a - x}{a - b}f(b) \\ f(x) - f(a) &\geq \frac{x - a}{b - a}f(b) - \frac{x - a}{b - a}f(a) \\ \frac{f(x) - f(a)}{x - a} &\geq \frac{f(b) - f(a)}{b - a} \end{aligned}$$

And similarly for the right two inequality terms.

Part (b): For $f(x) = -x^2 + 2x + 5$, $a = 0$, and $b = 3$ and $x = 1$:



Problem 2

Show that the maximum of a convex function f over the polyhedron $\mathcal{P} = \mathbf{conv}\{v_1, v_2, \dots, v_k\}$ is achieved at one of the vertices, that is

$$\sup_{x \in \mathcal{P}} f(x) = \max_{i=1,2,\dots,k} f(v_i).$$

In particular, let $f(x) = x_1^2 + 2x_1x_2 + 3x_2^2$, and let $v_1 = (1, 1)^T$, $v_2 = (1, -1)^T$, $v_3 = (-1, 1)^T$, and $v_4 = (-1, -1)^T$. Compute its maximum over the polyhedron.

Solution

From the text, we know that the convex hull of a set is the set of all convex combinations of points in that set. We can express $\mathcal{P}$ as

$$\mathcal{P} = \{\theta_1 v_1 + \dots + \theta_k v_k \mid \theta_1 + \dots + \theta_k = 1, \theta_i \geq 0, i = 1, \dots, k\}.$$

Thus, since any point $x \in \mathcal{P}$ can be expressed as a convex combination of the vertices $v_1, v_2, \dots, v_k$, we have

$$f(x) = f\left(\sum_{i=1}^k \theta_i v_i\right)$$

Since f is a convex function, by Jensen's inequality,

$$f\left(\sum_{i=1}^k \theta_i v_i\right) \leq \sum_{i=1}^k \theta_i f(v_i)$$

This value is maximized when $f(v_i)$ is maximized. Thus,

$$f(x) \leq \max_{v_i \in \mathcal{P}} f(v_i) \sum_{i=1}^k \theta_i \implies \sup_{x \in \mathcal{P}} f(x) = \max_{i=1,2,\dots,k} f(v_i).$$

Now we have $f(x) = x_1^2 + 2x_1x_2 + 3x_2^2$, and $v_1 = (1, 1)^T$, $v_2 = (1, -1)^T$, $v_3 = (-1, 1)^T$, and $v_4 = (-1, -1)^T$. We can compute

$$\begin{aligned} f(v_1) &= f((1, 1)^T) = 6, & f(v_2) &= f((1, -1)^T) = 2, \\ f(v_3) &= f((-1, 1)^T) = 2, & f(v_4) &= f((-1, -1)^T) = 6. \end{aligned}$$

Thus, the maximum over the polyhedron is 6.

Problem 3

Suppose

$$f(x) = h(g_1(x), g_2(x)),$$

where $g_1(x) = \|x\|_2^2$, $g_2(x) = \|x\|_1$, and $h(u, v) = \log(e^u + e^v)$.

- (a) Show $f(x)$ is convex by applying the correct composition rule.
- (b) Give another h that keeps f convex but is non-logarithmic, and explain why.

Solution

Part (a): h presents a log-sum-exp form, which is convex when both g_1 and g_2 are convex. Since both g_1 and g_2 are convex (all norms are convex), we can conclude that $f(x)$ is convex.

Part (b): Very many functions work here. Simply $h(u, v) = u + v$ works, as the sum of two convex functions is also convex.

Problem 4

Let $f_1, f_2, \dots, f_m$ be convex function on $\mathbb{R}^n$. Define

$$f(x) = \inf \{f_1(x_1) + f_2(x_2) + \dots + f_m(x_m) \mid x_i \in \mathbb{R}^n, x_1 + x_2 + \dots + x_m = x\}.$$

Show $f(x)$ is convex on $\mathbb{R}^n$.

Solution

Since each $f_1, f_2, \dots, f_m$ is convex, we have

$$f_i(\theta x_i + (1 - \theta)y_i) \leq \theta f_i(x_i) + (1 - \theta)f_i(y_i)$$

Thus, for our main function,

$$f(\theta x + (1 - \theta)y) \leq \sum_{i=1}^m (\theta f_i(x_i) + (1 - \theta)f_i(y_i))$$

And taking the infimum over all x_i and y_i preserves the inequality, so

$$f(x) = \inf \{f_1(x_1) + f_2(x_2) + \dots + f_m(x_m) \mid x_i \in \mathbb{R}^n, x_1 + x_2 + \dots + x_m = x\}$$

is convex.

Problem 5

The r -separation function $\Delta_r(x)$ of the vector $x \in \mathbb{R}^n$, where $r \leq n$, is defined as the difference between the sum of the r largest components of the vector x and the sum of the r smallest components.

- (a) Is the function $\Delta_r(x)$ convex? Prove your answer.
- (b) Let $r = 1$ and $n = 2$. Sketch the function $\Delta_1(x)$ in the positive quadrant.

Solution

Part (a): The function $\Delta_r(x)$ written out:

$$\Delta_r(x) = \sum_{r \text{ largest}} x_i - \sum_{r \text{ smallest}} x_i$$

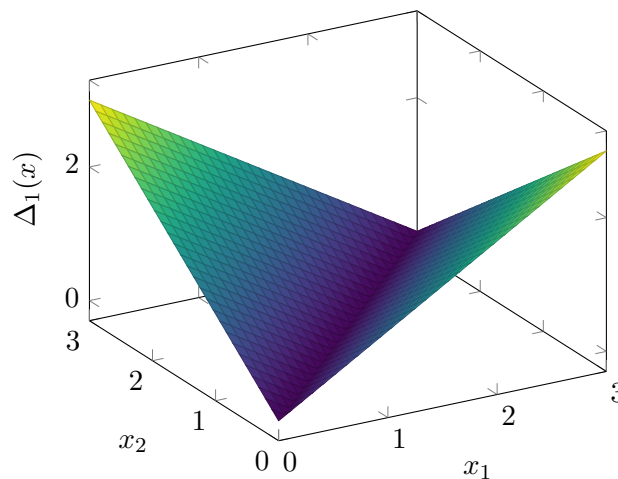
We know from the notes that the sum of r largest components of $x \in \mathbb{R}^n$ is convex. It is obvious when presented as

$$f(x) = \max\{x_{i_1} + x_{i_2} + \dots + x_{i_r} \mid 1 \leq i_1 < i_2 < \dots < i_r \leq n\}$$

and the r smallest components is concave, but the minus sign makes it convex.

Since we are adding $\Delta_r(x) = f(x) + (-g(x))$, where both f and $-g$ are convex, we can conclude that $\Delta_r(x)$ is convex, by the convex summation property.

Part (b): For $r = 1$ and $n = 2$,



Problem 6

Suppose $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is convex, and $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, and $d \in \mathbb{R}$. Show that the function that $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, where

$$g(x) = (c^T x + d)f\left(\frac{Ax + b}{c^T x + d}\right), \quad \text{dom } g = \{x \mid c^T x + d > 0\}.$$

Provide a clear argument which rules you are applying, and the exact manner of function composition.

Solution

We have the perspective of some function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined as $g : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$,

$$g(x, t) = tf(x/t), \quad \text{dom } g = \{(x, t) \mid x/t \in \text{dom } f, t > 0\}.$$

g is convex if f is convex.

Here, $x = Ax + b$ and $t = c^T x + d$, which is affine. Since convexity is preserved during composition with affine functions, g is also convex.

Problem 7

A vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called monotone if $\langle f(x) - f(y), x - y \rangle \geq 0$. Show that the gradient of a convex function is monotone.

Solution

If the gradient of a convex function is monotone, then

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$$

By the 1st-order condition,

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) = f(x) + \langle \nabla f(x), y - x \rangle$$

and

$$f(x) \geq f(y) + \nabla f(y)^T(x - y) = f(y) + \langle \nabla f(y), x - y \rangle$$

Adding these two, we get

$$\begin{aligned} f(x) + f(y) &\geq f(x) + f(y) + \langle \nabla f(x), y - x \rangle + \langle \nabla f(y), x - y \rangle \\ 0 &\geq \langle \nabla f(x), y - x \rangle + \langle \nabla f(y), x - y \rangle \\ &= \langle \nabla f(x) - \nabla f(y), y - x \rangle \\ &= -\langle \nabla f(x) - \nabla f(y), x - y \rangle \end{aligned}$$

Thus,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$$

proving that the gradient of a convex function is monotone.

Problem 8

The proximal operator of a function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ is defined by

$$\text{prox}_{\lambda f}(v) = \arg \min_x \left(f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right).$$

- (a) Is the proximal operator a convex function?
- (b) Derive the proximal operator for $f(x) = \|x\|_1$.
- (c) Solve explicitly for $v = (3, -2, 0.5)^T$ and $\lambda = 1$ when $f(x) = \|x\|_1$.

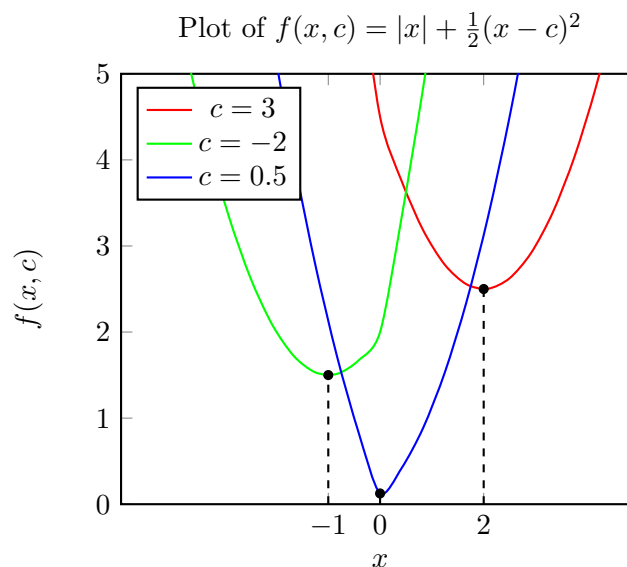
Solution

Part (a): Since $\nabla^2 p \not\leq 0$, the proximal operator is not convex.

Part (b): Now $f(x) = \|x\|_1$. Thus, we have

$$\begin{aligned} \text{prox}_{\lambda f}(v) &= \arg \min_x \left(\|x\|_1 + \frac{1}{2\lambda} \|x - v\|_2^2 \right) \\ &= 1/\lambda (x - v) \end{aligned}$$

Part (c): Here $v = (3, -2, 0.5)^T$, $\lambda = 1$, and $f(x) = \|x\|_1$. I'll plot each dimension of the function to see its minimum.



From the graph, we can see that the minimums are at $x = 2$, $x = -1$, and $x = 0$.

Problem 9

Given a norm $\|\cdot\|$ on $\mathbb{R}^n$, the dual norm is defined as

$$\|y\|_* = \sup_{\|x\| \leq 1} y^T x.$$

- (a) Show that $\|\cdot\|_*$ is indeed a norm.
- (b) Compute the dual norm of $\|x\|_1$ and $\|x\|_\infty$.
- (c) For $x = (1, 2)^T$, compute $\|x\|_2^*$.

Solution

Part (a): To be a norm it must satisfy the triangle inequality

$$\|y_1 + y_2\|_* = \|y_1\|_* + \|y_2\|_* \quad \rightarrow \quad \sup_{\|x\| \leq 1} (y_1 + y_2)^T x \leq \sup_{\|x\| \leq 1} y_1^T x + \sup_{\|x\| \leq 1} y_2^T x$$

which holds by the properties of supremum.

It must also satisfy absolute homogeneity

$$\|sy\|_* = |s| \|y\|_* \quad \rightarrow \quad \sup_{\|x\| \leq 1} (sy)^T x = |s| \sup_{\|x\| \leq 1} y^T x$$

this condition also holds since $\sup(sy) = s \sup(y)$.

Finally, non-negativity, which is also satisfied by the norm definition.

Part (b): For $\|x\|_1$, the dual norm is simply the l_∞ norm and vice versa.

Part (c): For $x = (1, 2)^T$, $\|x\|_2^* = \|x\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}$.